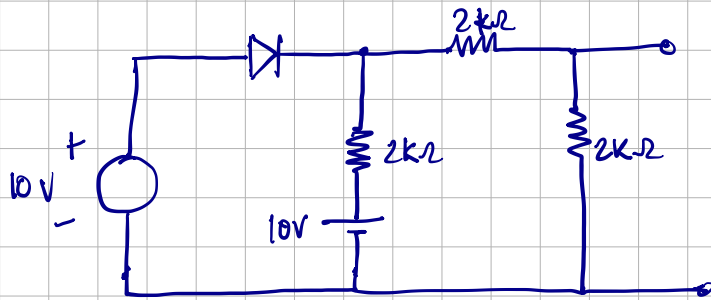


WEEK-6 GRADED ASSIGNMENT

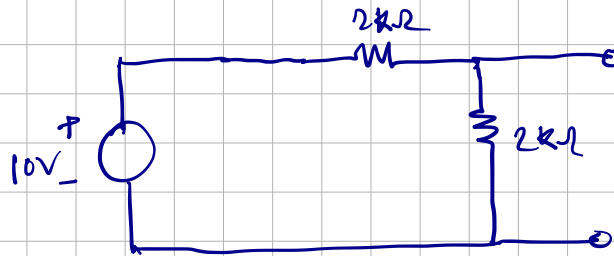
NAME:- SHANU ROHIT KAMLESH

Roll No:- 25F3100145

Q.1]
Soln:-

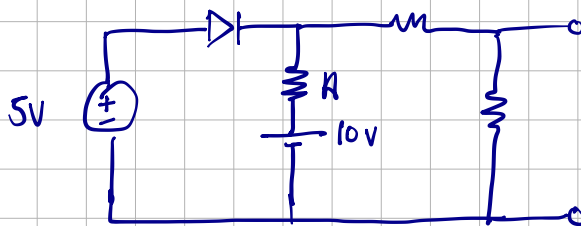


In this case diode will be forward bias:-



$$\therefore V_0 = \frac{10 \times 2k\Omega}{4k\Omega} = 5V$$

Case-2



maximum voltage drop across A resistor.

$$\frac{10}{6} \times 2 = \frac{10}{3}$$

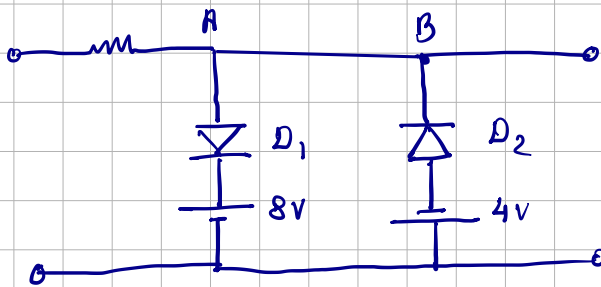
Hence cathode voltage will always be greater than Anode voltage. Hence diode will be off

$$\therefore \therefore \text{Voltage across load} = \frac{10}{6} \times 2$$

$$= \frac{10}{3} = 3.34$$

$$\therefore \frac{V_{01}}{V_{02}} = \frac{5 \times 3}{10} = \frac{3}{2} \text{ V}$$

Q.2]
Soln:-

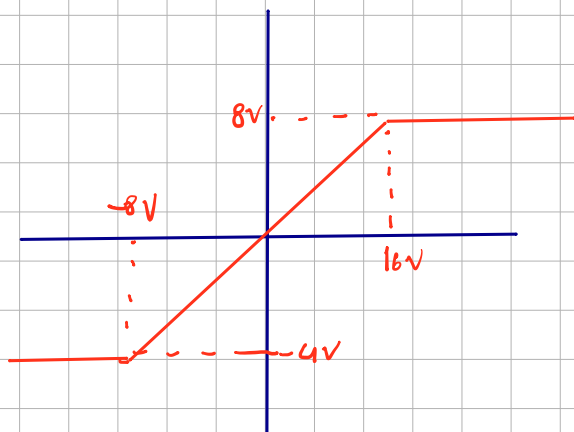


If $V_A > 8\text{V}$ D_1 will be forward bias

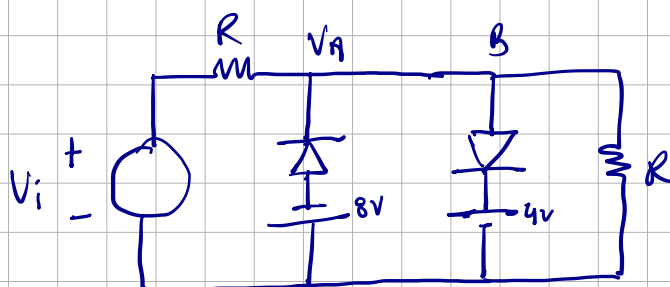
If $V_B < -4\text{V}$ D_2 will be forward bias.

Hence between 8V to -4V both diode will be reverse biased as we will get applied voltage across output.

Q.3)

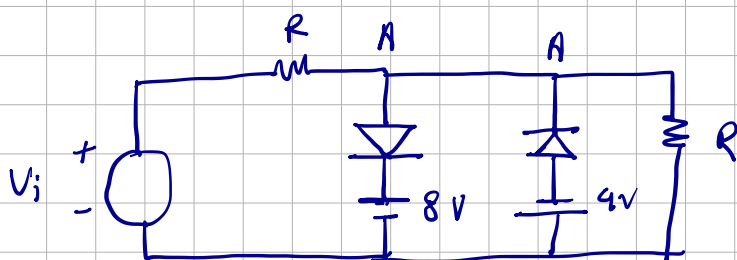


As per diagram both diodes must be off between $-8V$ to $16V$



here $V_B > 4V$ to be in forward bias mode.

It means after $8V$ output voltage will be constant



In this circuit If V_A is greater than $8V$ the D_1 will be f.b & D_2 will be reverse bias.

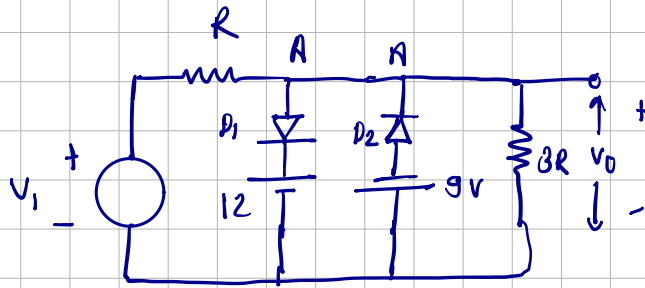
$\therefore V_o = 16V$ as Both resistance are of same value.

Similarly If $V_A < -4V$ then D_2 will be forward bias.

"Hence option B is correct"

Q.4]

Soln:-



$\therefore D_1$ will be forward biased when $V_A > 12V$

D_2 will be forward biased when $V_A < -9V$

$$\therefore V_A = \frac{V_i \cdot 3R}{4R} = \frac{3}{4} V_i$$

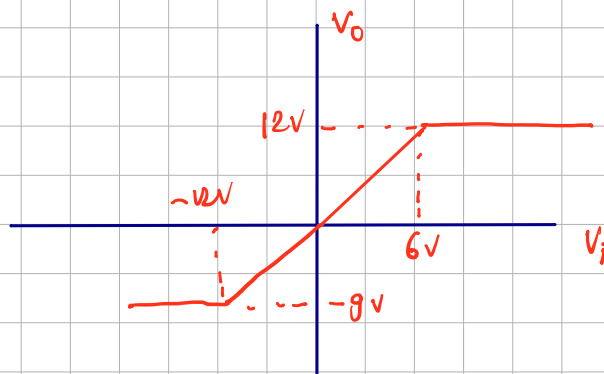
$$\frac{3}{4} V_i > 12V \quad \& \quad \frac{3}{4} V_i < -9V$$

$$V_i > 16V \quad \& \quad V_i < -12V$$

$\therefore$ As $V_i > 16V$ output will be constant to $12V$

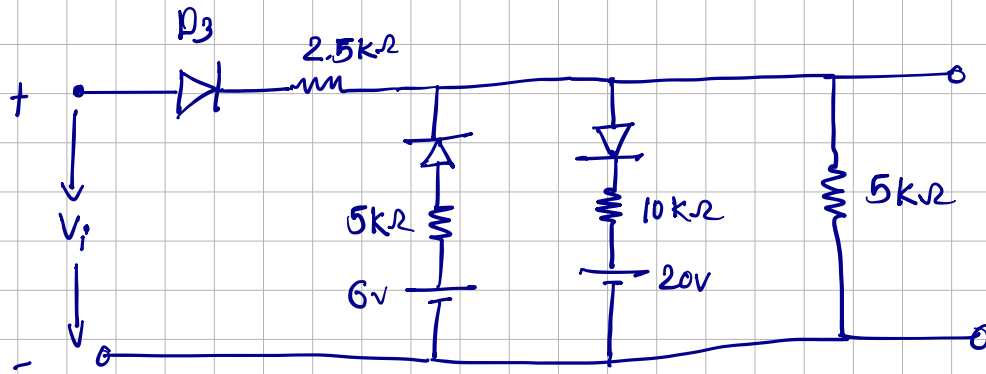
As $V_i < -12V$ output will be constant to $-9V$

"Hence option 1 is correct."



Q.5]

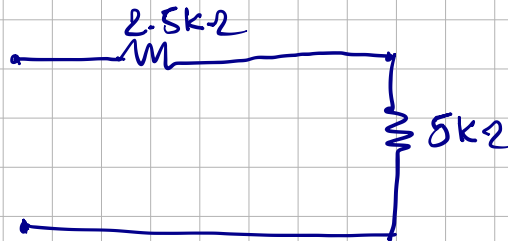
Soln:-



Diode D_3 will always be forward bias.

Let say D_3 is forward bias &

D_2 & D_1 are reverse biased.



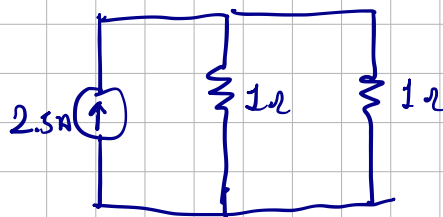
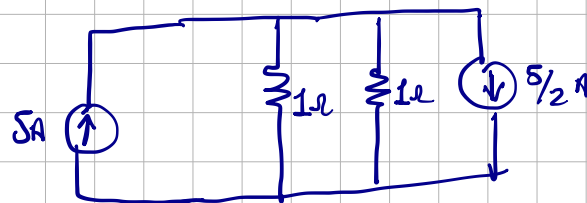
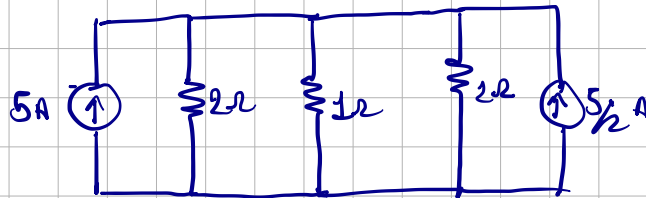
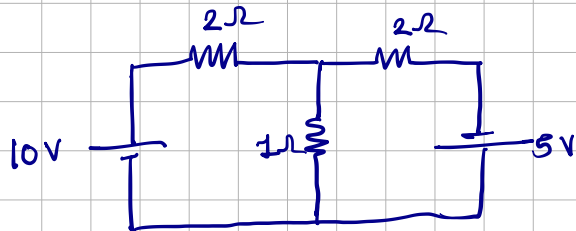
$$\therefore V_5 = \frac{V_i \times 5}{7.5}$$

$$V_5 = \frac{2V_i}{3}$$

$$V_{out} = \frac{2}{3} V_i$$

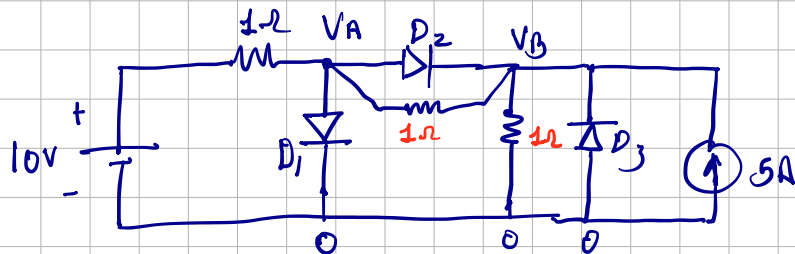
$$\therefore \text{Slope} = 0.66$$

Q6]



$$\therefore V_0 = \frac{2.5}{2} = 1.25 \text{ V}$$

Q7]

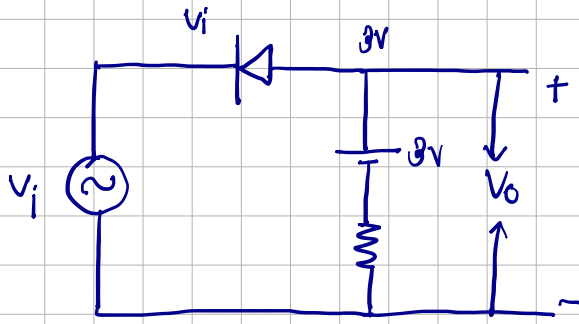


$D_1 \rightarrow \text{ON}$

$D_2 \rightarrow \text{OFF}$

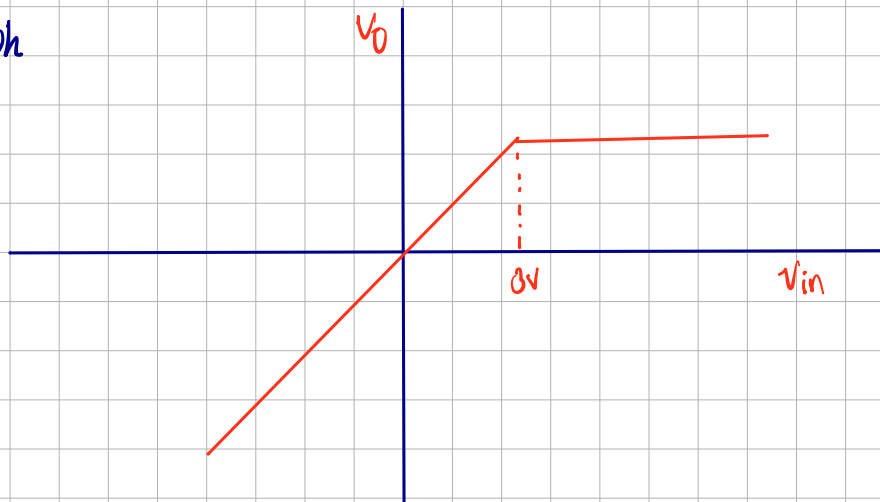
$D_3 \rightarrow \text{OFF}$

Q.8)



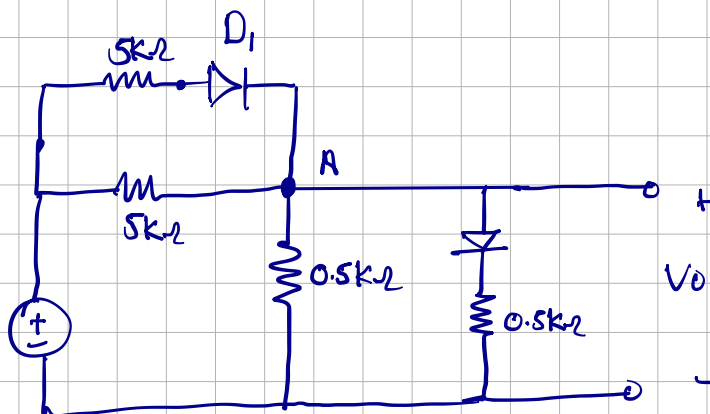
$\Rightarrow$ $V_i < 3V$ for diode to be in F.B
 $\text{If } V_i > 3V$ then diode will be R.B
 & hence $V_o = 3V$

Graph



Q.9]

Soln :-



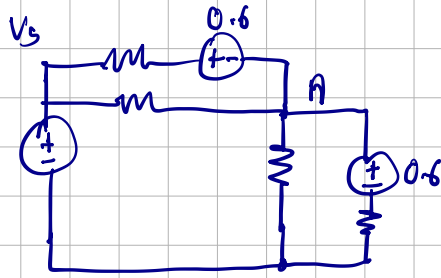
For both diode to be forward bias

$$V_A > 0.6$$

$$V_{D_1} > V_A + 0.6$$

$$V_{D1} > 1.2$$

now



$$\frac{V_S - V_A}{5k} + \frac{V_S - (V_A + 0.6)}{5k} = \frac{V_A}{0.5k} + \frac{V_A - 0.6}{0.5k}$$

$$V_A = \frac{V_S + 2.7}{11}$$

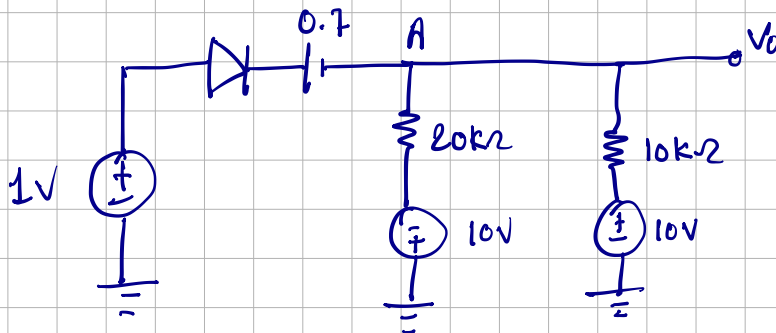
$$V_A > 0.6$$

$$\therefore \frac{V_S + 2.7}{11} > 0.6$$

$$V_S > 6.6 - 2.7$$

$$V_S > 3.9 \text{ V}$$

Q.10)



D_1 will be forward biased.

$\therefore$ 0.3 V will be as V_A